BASIC CIVIL & MECHANICAL ENGINEERING

Subject Code: 23A01201T | UNIT – III : TRANSPORTATION, WATER RESOURCES & ENVIRONMENTAL ENGINEERING

I. Importance of Transportation in a Nation's Economic Development

> **Transportation** = movement of people and goods from one place to another. It is the **lifeline of a nation's economy.**

- **Modes:** roads (highways), railways, waterways (inland, sea), airways, pipelines, ropeways.

❖Importance

- Links **production centres, markets, raw material sources** → trade & industrial growth.
- Provides **employment** (drivers, rail/ port staff, construction) and promotes tourism.
- **Integrates** the country socially & politically; access to health, education, remote & border areas.
- Reduces **cost of goods**, transport cost, time; gives **price stability**, balanced regional development (agriculture → mandi/ export).
- Essential for **defence**, disaster relief, urban development, GDP growth (transport ≈ 6% of GDP).
- Indian scenario: road network ≈ **6.3 million km** (2nd largest in the world); National Highways ≈ 1.46 lakh km (Bharatmala, Golden Quadrilateral, NHAI); railways ≈ 68,000 route km (Indian Railways – 4th largest); 13 major + 200 minor ports (Sagarmala); 150+ airports (UDAN); metro rail in 20+ cities.

❖Road classification (Nagpur Plan)

- National Highways (NH), State Highways (SH), Major District Roads (MDR), Other District Roads (ODR), Village Roads (VR); also expressways.

❖Components of a highway

- Carriageway, shoulders, median, kerb, side drains, camber (1 in 50), embankment, pavement, culverts/ bridges, road signs & markings.

2. Types of Highway Pavements

> Pavement = durable surface material laid over the subgrade to carry traffic loads safely and distribute them to the soil beneath. Two types: Flexible & Rigid (and composite).

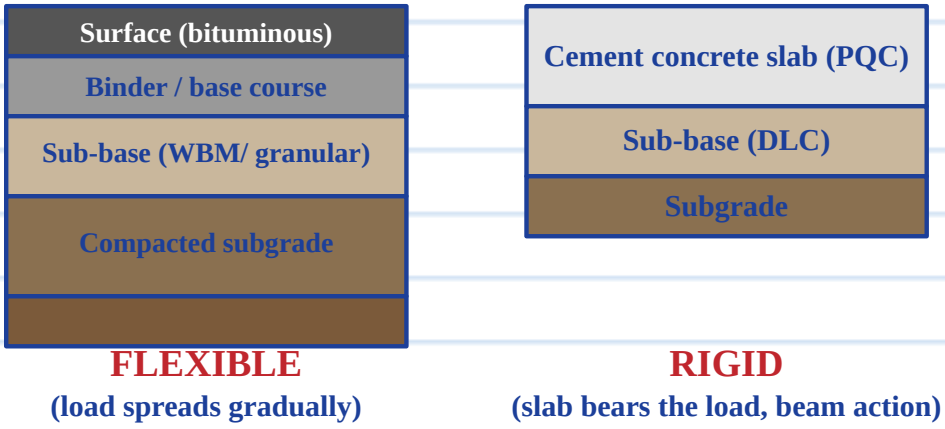


Fig 3.1 Flexible vs rigid pavement composition

❖2.1 Flexible Pavements

- Made of **bituminous (asphalt) surface** over granular layers; **load is transferred by grain-to-grain contact**, spreading in a **cone shape** and decreasing with depth; each layer shows **small deflection** under wheel load (flexes).
- Layers (top → bottom):
  - **Surface/ wearing course** – bituminous concrete (BC), semi-dense BC, mastic asphalt – resists wear, waterproof.
  - **Binder course** – dense bituminous macadam (DBM).
  - **Base course** – WBM (water bound macadam), WMM (wet mix macadam).
  - **Sub-base course** – granular (GSB, moorum, gravel).
  - **Subgrade** – compacted natural soil (CBR-based design).
- Design method: **CBR method (IRC 37)**, group-index method.
- Life: 10–15 years; **initial cost low**, maintenance high (potholes, rutting, bleeding).

❖2.2 Rigid Pavements

- Cement concrete slab (PQC – pavement quality concrete M30–M40, thickness 200–300 mm) over a dry lean concrete (DLC) sub-base; the stiff slab acts as a beam, spreading load over a large area of subgrade (beam action).
- Joints (expansion, contraction, longitudinal, construction) with dowel bars & tie bars to control cracking.
- Types: Plain concrete (JPCP), Jointed reinforced (JRCP), Continuously reinforced (CRCP), Prestressed.
- Design: Westergaard's theory, IRC 58. Life: 30–40 years; initial cost high, maintenance low.

❖Simple differences – Flexible vs Rigid

Point	Flexible	Rigid
Surface	Bituminous	Cement concrete slab
Load transfer	Grain to grain, layer by layer	Slab bending (beam action)
Design	Thickness by CBR	Slab thickness by flexural strength
Deformation	Takes deformation (settlement)	Rigid, may crack
Joints	No joints	Joints required
Initial cost	Low	High
Maintenance	High, frequent	Low
Life	10–15 yr	30–40 yr
Night visibility	Poor (dark)	Good (light colour)
Fuel/ temp	Affected by heat (melts)	Heat resistant
Repair	Easy, quick opening	Difficult
Stage construction	Possible	Not possible

3. Basics of Harbour Engineering

> A harbour is a sheltered area of water where ships can anchor/ berth safely for loading, unloading, repairs and refuge from storms.

- Classification: Natural (e.g., Mumbai, Kochi), Semi-natural (protected by break-water on one side), Artificial (fully protected by break-waters – Chennai); by use – Commercial, Refuge, Fishery, Naval, Military.

- **Components:** break-waters (protect from waves), jetties, wharves, quays, piers (berthing structures), dolphins, dredged channels, docks (wet dock with lock gate, dry dock for repair), navigation aids (lighthouse, buoys), transit sheds, warehouses, cranes, container yard, fenders (absorb ship impact).
- **Site selection:** sufficient water depth (≥ 10–15 m), natural protection, good anchorage, hinterland connectivity (rail/ road), low siltation, availability of land.
- **Terms:** Draft (depth of ship under water), tide, tidal range, wave action, siltation, dredging.
- **Major ports of India:** Mumbai, JNPT, Kandla (Deendayal), Mormugao, New Mangalore, Kochi, Tuticorin (V.O. Chidambaranar), Chennai, Ennore (Kamarajar), Visakhapatnam, Paradip, Kolkata (Haldia). Sagarmala project.

4. Basics of Tunnel Engineering

> **Tunnel** = an underground passage (horizontal/ slightly inclined) driven through hill/ soil/ under river without disturbing the surface; length > 100 m, length > width.

- **Purposes:** railways/ roads through mountains, metro, water conveyance (hydropower, irrigation), sewers, utilities, mines, defence.
- **Advantages:** shorter route, saves time & fuel, avoids traffic and land acquisition, protected from weather.

❖Classification

- By alignment (horizontal/ vertical/ inclined), by material (soil, hard rock), by position: hill tunnel, subaqueous (river), urban/ cut-and-cover tunnel.
- Shapes: circular, horse-shoe (D-shape), egg-shaped, rectangular.

❖Methods of tunnelling

- Drill & blast (hard rock) → mucking (removal) → support (steel ribs, shotcrete, rock bolts) → lining (concrete). NATM (New Austrian Tunnelling Method).
- TBM (Tunnel Boring Machine) – full-face mechanical excavation, used for metro/ soft ground.

- Cut and cover – open trench roofed over (shallow metro).
- Immersed tube for river crossings.
- Needs ventilation, drainage, lighting and safety.
- Examples in India: Atal Tunnel (Rohtang, 9.02 km, Himachal), Pir Panjal rail tunnel (J&K), Konkan Railway tunnels, Delhi Metro, Chenani–Nashri tunnel, Banihal.

5. Basics of Airport Engineering

> Airport = a facility where aircraft take off, land, are parked, serviced, and passengers/ cargo are handled.

- Components: Runway (take-off/ landing strip, oriented along prevailing wind; length 1000–4000 m), taxiway (connects runway & apron), apron (parking & loading area), terminal building, control tower (ATC), hangars, navigation aids (ILS, VOR), lighting, windsock, parking, cargo & fuel stations.
- Site selection: away from city (noise) but accessible; flat, well-drained land; clear approach zones (no obstructions); good soil; wind direction; meteorology (fog); room for expansion.
- Runway orientation: along prevailing wind (wind rose, wind coverage ≥ 95%); aircraft takes off/ lands against wind.
- Runway length depends on aircraft type, altitude (↑ height → ↑ length), temperature, gradient, wind: correction for elevation, temperature, gradient.
- Classification (ICAO): Code A–F by wingspan (A: < 15 m... F: 65–80 m); International, domestic; Greenfield airports (Navi Mumbai, Noida, Bhogapuram), regional – UDAN scheme.
- Pavements: flexible (asphalt), rigid (concrete, heavy aircraft).
- Zones: approach, turning, transition, horizontal and conical surfaces (imaginary surfaces for obstruction clearance).

6. Basics of Railway Engineering

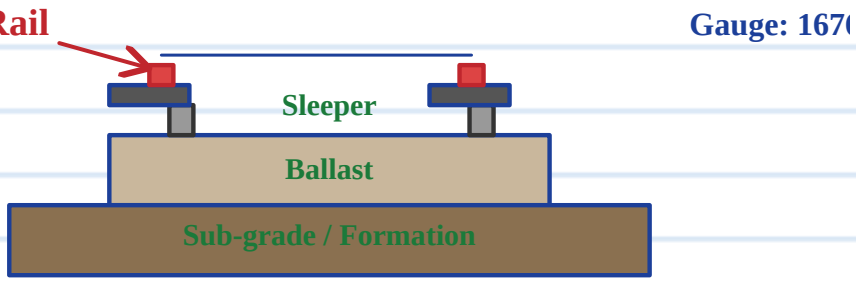


Fig 3.2 Railway track cross-section (Indian Broad Gauge = 1676 mm)

> **Railway** = guided transport on a **permanent way (track)** with steel rails, carrying trains hauled by locomotives; cheapest for **heavy and long-distance** traffic.

❖Components of a track (permanent way)

- **Rails:** steel (52 kg/m, 60 kg/m, 90 UTS), flat-footed type; carry loads & guide wheels.
- **Sleepers:** transverse supports under rails – **concrete (PSC)** most common, steel, wooden, cast-iron; spacing ~ 60 cm (1660 per km). They hold gauge, distribute load, insulate.
- **Fasteners:** fish plates/ elastic clips/ pandrol clips, **bolts, spikes, base plates.**
- **Ballast:** crushed stone (20–50 mm) 250–350 mm thick under sleepers – drains, resilient, distributes load, holds track in position.
- **Formation/ Subgrade** and embankment/ cutting.

❖Gauge (distance between inner faces of rails)

Gauge	Width	Where
Broad Gauge (BG)	1676 mm	Main Indian lines (~ 96%)
Metre gauge (MG)	1000 mm	Few branch lines
Narrow gauge (NG)	762 / 610 mm	Hill railways (Darjeeling, Nilgiri, Kalka-Shimla)
Standard gauge	1435 mm	Metro, world, high-speed

- **Other topics: points & crossings (turnouts)** to shift trains from one track to another; **signals, interlocking;** stations & yards; **gradient, super-elevation (cant)** on curves; **coning of wheels (1 in 20); creep.**

- Modern: Electrification (25 kV AC), Vande Bharat, Dedicated Freight Corridors (DFC), Metro, Mono, bullet train (MAHSR).

7. WATER RESOURCES & ENVIRONMENTAL ENGINEERING – Introduction

- Water covers 71% of earth but only 2.5% is fresh, and < 1% is easily accessible (lakes, rivers, groundwater).
- India: 16% of world population, 4% of water resources; average annual rainfall 1170 mm; monsoon (June–Sept) gives 75% of rain → need storage & management.
- Water resources engineering – assessment, planning, development, utilisation & control of water for drinking, irrigation, power, industry, navigation, flood control.
- Environmental engineering – provides safe water, sanitation, pollution control.
- Water demand (urban): 135 LPCD (litres per capita per day) for domestic (CPHEEO), 40–70 LPCD in rural; includes domestic, industrial, public, fire, losses.

8. Sources of Water

- Surface sources: rivers, streams, lakes, ponds, reservoirs (impounding), canals – turbid, bacteria, easily polluted, large quantity, needs treatment.
- Ground water sources: wells (open/ dug, tube wells, bore wells), springs, infiltration galleries – clear, low turbidity, hard (more minerals), usually safe, limited yield; recharge by rain.
- Rain water: collected from roofs; pure but acidic, needs storage.
- Others: sea water (desalination), recycled waste-water.

❖Surface vs underground source

Feature	Surface	Ground
Turbidity	High	Low
Bacteria	More	Less
Hardness	Soft	Hard
Quantity	Large	Limited
Treatment	Full (coagulation, filtration)	Usually disinfection only

9. Quality of Water & Specifications

> Potable water = water that is safe to drink – clear, colourless, odourless, tasteless, free from pathogens, toxic substances, and palatable. Standard: IS 10500 : 2012 (BIS drinking water specification); also WHO guidelines.

❖Impurities

- Suspended (silt, clay, algae), colloidal, dissolved (salts, gases, organics), bacteria/ viruses/ protozoa (cholera, typhoid, hepatitis, diarrhoea).

❖Water quality parameters

- Physical: turbidity (NTU), colour (Hazen units), taste & odour, temperature, total solids/ TDS, conductivity.
- Chemical: pH, hardness (Ca, Mg – $CaCO_3$ mg/L), alkalinity, chlorides, sulphates, fluoride, nitrates, iron, manganese, arsenic, heavy metals, dissolved oxygen (DO), BOD, COD.
- Biological/ bacteriological: coliform bacteria (E. coli) – indicator; MPN/ 100 mL.

❖Drinking-water limits (IS 10500, desirable / permissible)

Parameter	Acceptable	Permissible (no alternate)
pH	6.5 – 8.5	no relaxation
Turbidity	1 NTU	5 NTU
TDS	500 mg/L	2000 mg/L
Total hardness ($CaCO_3$)	200 mg/L	600 mg/L
Chloride	250 mg/L	1000 mg/L
Sulphate	200 mg/L	400 mg/L
Fluoride	1.0 mg/L	1.5 mg/L
Nitrate	45 mg/L	no relaxation
Iron	0.3 mg/L	no relaxation (1.0)
Arsenic	0.01 mg/L	0.05 mg/L
Residual chlorine	0.2 mg/L	1.0 mg/L
Coliform / E. coli	Nil / 100 mL	Nil

- Effects: excess fluoride → fluorosis; nitrate → blue-baby disease; arsenic → cancers; hardness → scale & soap waste.

❖Water treatment (brief flow)

Raw water → Screening → Sedimentation (+ coagulation: alum) → Filtration (rapid/ slow sand) → Disinfection (chlorination, UV, ozone) → Storage → Distribution

- Aeration removes gases/ iron; softening removes hardness (lime-soda, ion exchange); RO for TDS; disinfection with chlorine (0.2–0.5 mg/L residual).
- Waste-water (sewage) treatment: preliminary (screens, grit) → primary (sedimentation) → secondary (biological – activated sludge, trickling filter) → tertiary → disinfection; sludge digestion.

10. Introduction to Hydrology

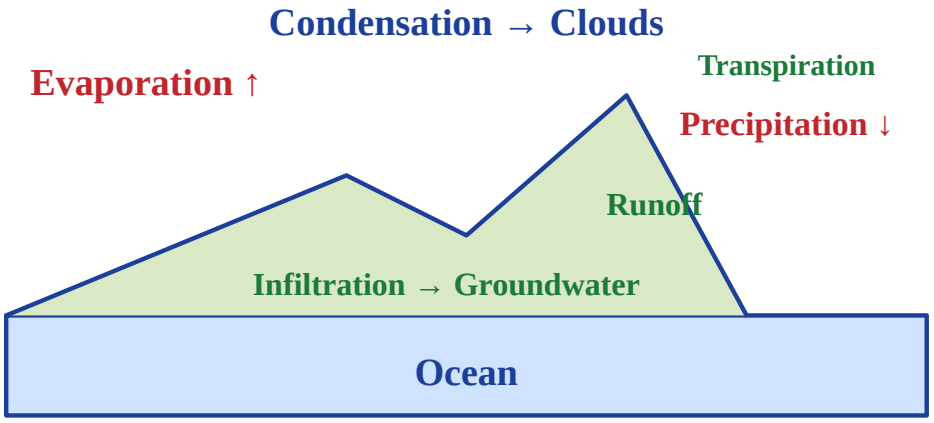


Fig 3.3 Hydrologic cycle

> Hydrology is the science dealing with the occurrence, circulation, distribution and properties of water on earth – above, on and below the surface.

❖Hydrologic cycle

- Evaporation (sun heats oceans/ lakes) + transpiration (plants) = evapotranspiration → condensation (clouds) → precipitation (rain, snow, hail, dew) → interception, infiltration (into soil), percolation (to ground water), surface runoff (streams) → rivers → ocean.
- Water balance equation: $P = R + E + I + \Delta S$ (Precipitation = Runoff + Evapotranspiration + Infiltration + change in storage).

❖Precipitation

- Forms: rainfall, snow, sleet, hail, drizzle, dew, frost.

- Types of rainfall: convective (heating of ground – tropical), orographic (air forced up mountains – Western Ghats, North-east), cyclonic (low-pressure; frontal & non-frontal – coastal A.P., Odisha).
- Measurement: rain gauge – non-recording (Symon's gauge, 127 mm/ 5" funnel) and recording (tipping-bucket, float, weighing); unit mm; rain gauge density 1 per 520 km^2 (plain).
- Average rainfall over a catchment: arithmetic mean, Thiessen polygon method, isohyetal method.

Arithmetic mean = $(P_1+P_2+...+P_n)/n$; Thiessen mean = $\Sigma(P_iA_i)/\Sigma A_i$

❖Runoff

- Runoff = part of rainfall that flows over land to streams. Depends on rainfall intensity/ duration, catchment area, slope, soil, land use, vegetation, urbanisation.
- Rational formula (peak discharge): $Q = (1/360)\ C\ i\ A$ (Q in m^3/s, i in mm/h, A in hectares, C = runoff coefficient 0.1 – 0.95).
- Hydrograph (discharge vs time), catchment/ watershed, stream gauging (current meter), infiltration indices (φ-index).
- Flood, drought, base flow.

❖Example

Catchment A = 50 ha, C = 0.4, i = 60 mm/h → $Q = (1/360) \times 0.4 \times 60 \times 50 = 3.33\ m^3/s$.

Gauges P_1 = 80, P_2 = 100, P_3 = 120 mm → arithmetic mean = 100 mm.

II. Rainwater Harvesting (RWH)

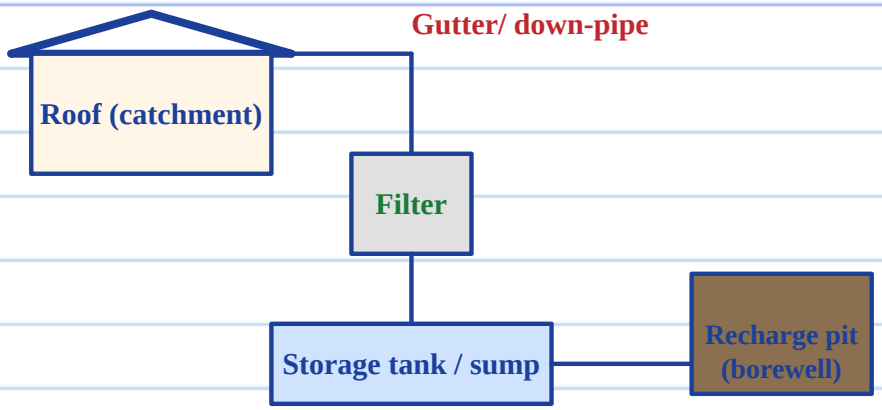


Fig 3.5 Rooftop rainwater harvesting system

> Rainwater harvesting = collecting, storing and using rain water (or recharging groundwater) where it falls, instead of letting it run off.

❖Need

- Falling water tables (over-exploitation), shortage in urban areas, floods & runoff, poor groundwater quality, increases self-sufficiency & saves energy of pumping.

❖Methods

- (a) Rooftop harvesting – storage: roof water → gutters → down pipes → first-flush diverter → filter (sand/ gravel/ charcoal) → underground sump/ overhead tank for direct use.
- (b) Recharge methods: recharge pit/ trench, recharge shaft, recharge through borewell/ dug well, percolation tank, check dams, gully plugs, contour bunds, farm ponds; injection wells for deep aquifers.
- Traditional systems: johads, kunds, baoli/ stepwells, tankas (Rajasthan), ahar-pynes, eris (Tamil Nadu), katas, bhandaras, cheruvu (A.P.).

❖Components of a roof-top system

- Catchment (roof), gutters & conduits, first-flush device (discard first 1 mm of rain), filter unit, storage tank or recharge structure, overflow.
- Design: Annual harvestable volume $V = A \times R \times C$ (A = roof area, R = rainfall, C = coefficient ≈ 0.8–0.85).

❖Example

> Roof 100 m^2, annual rain 800 mm = 0.8 m, C = 0.8: $V = 100 \times 0.8 \times 0.8 = 64\ m^3 = 64{,}000$ L/year.

❖Benefits

- Increases ground water table, reduces runoff/ flooding, improves water quality (dilutes fluoride/ salts), cheap, easy to maintain, reduces dependence on piped supply.
- Made mandatory in many cities (A.P. – "Neeru-Chettu", Hyderabad Water Board, Chennai).

12. Water Storage & Conveyance Structures

❖Storage structures

- Dams & reservoirs, tanks (ponds), check dams, storage tanks (ground-level, elevated, underground sumps), service reservoirs, balancing reservoirs, water towers.
- Reservoir zones: Dead storage (below minimum level, silt) → Live/ useful storage (between MDDL & FRL) → Flood/ surcharge storage (above FRL up to MWL).

❖Conveyance structures

- Canals (lined/ unlined), pipelines (CI, DI, PVC, HDPE, RCC, steel), aqueducts, siphons, flumes, tunnels, culverts, pumps with rising mains, penstocks.
- Pipe flow – Manning/ Hazen–Williams formulas. $Q = A\,V$.
- Distribution systems: gravity, pumping, dual; layouts – grid-iron, ring, radial, dead-end; valves, hydrants, meters, water towers maintaining 7–15 m residual head.

13. Dams and Reservoirs – Simple Introduction

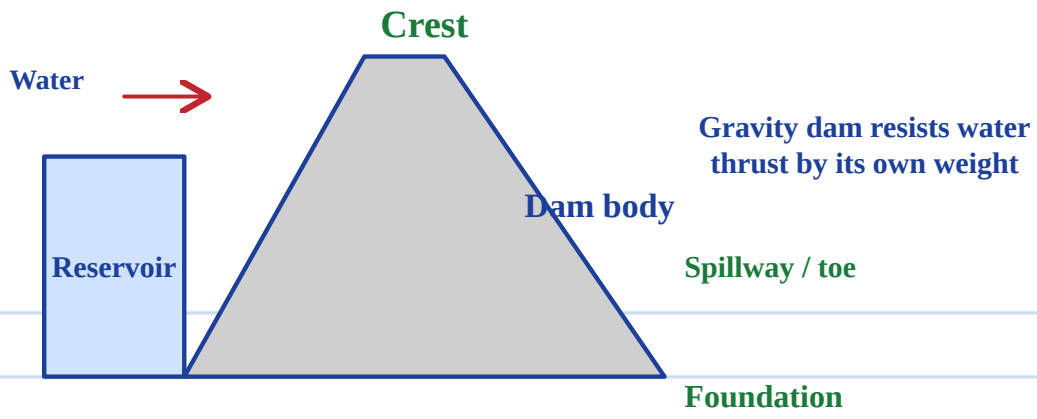


Fig 3.4 Gravity dam (profile) retaining a reservoir

A dam is a barrier constructed across a river to store water, forming a reservoir. India has ~5,700 large dams (third in world). Largest: Tehri (261 m, Uttarakhand), Bhakra (226 m), Hirakud, Nagarjuna Sagar (A.P.–Telangana, masonry), Srisailam, Polavaram, Sardar Sarovar, Krishnaraja Sagar.

❖Purposes

- Irrigation, hydroelectric power, drinking & industrial water supply, flood control, navigation, fisheries, recreation, groundwater recharge.

❖Classification

- By function: storage, diversion, detention, multipurpose.
- By material: masonry & concrete gravity, arch, buttress, earth-fill (embankment), rock-fill, steel/ timber.
- By structure/ hydraulic design: overflow (spillway) & non-overflow.
- Gravity dam: resists water thrust by own weight; needs good rock foundation; trapezoidal section; made of concrete/ masonry.
- Arch dam: curved in plan; transfers load to abutments; for narrow gorges (Idukki).
- Earth dam: local soil, impervious core, flatter slopes; for weak foundations; cheaper.

❖Components

- Crest, upstream/ downstream face, heel & toe, foundation, galleries, spillway (passes surplus flood safely), gates, outlets/ sluices, energy dissipators (stilling basin), fish ladder, power house (hydro dams).

❖Forces on a gravity dam

- Self weight (W), water pressure ($P = \frac{1}{2} \gamma_w H^2$ per metre), uplift pressure, silt pressure, wave, ice, earthquake, wind forces.
- Stability: safe against overturning, sliding, tension & crushing (no tension in base).

❖Reservoir

- Planning: capacity – elevation curve (from contour map), site selection (narrow gorge, wide valley, good rock, large catchment, low evaporation), sedimentation, submergence & rehabilitation (R & R).
- Levels: MDDL, FRL (full reservoir level), MWL (maximum water level); freeboard.

❖Advantages & problems

- Benefits: water, power, flood control. Problems: displacement, submergence of forests, siltation, seismicity, health (water-borne diseases), ecological downstream effects, high cost.

14. Solved Problems

❖Problem 1 – Water demand

> Population 20,000 × 135 LPCD = 2,700,000 L/day = **2700 m^3/day** = 31.25 L/s.

❖Problem 2 – Hardness

> Ca = 60 mg/L, Mg = 24 mg/L: hardness (as $CaCO_3$) = 2.5 Ca + 4.1 Mg = 150 + 98.4 = **248 mg/L** (hard; above 200 desirable).

❖Problem 3 – Hydrostatic force on dam

> Water depth H = 40 m: $P = ½ \times 9.81 \times 40^2$ = **7848 kN per metre**, acting at H/3 = **13.33 m** from base.

❖Problem 4 – Reservoir capacity

> Areas enclosed by contours at 100, 102, 104 m = 0.5, 1.2, 2.0 km^2 → V = (0.5+1.2)/2×2 + (1.2+2.0)/2×2 = 1.7 + 3.2 = **4.9 million m^3**.

15. Quick Revision & Important Questions

Topic	Key point
Highway pavements	Flexible = bitumen, grain-to-grain; Rigid = concrete slab, beam action
Railway	BG = 1676 mm; rails, sleepers, ballast
Water quality	pH 6.5–8.5; coliform nil; TDS ≤ 500 mg/L
Hydrology	$P = R + E + I + \Delta S$; $Q = CiA/360$
RWH	$V = A \times R \times C$
Dam	Gravity dam – self weight; crest, spillway, reservoir

- Explain the importance of transportation in economic development.
- Differentiate flexible and rigid pavements with sketches.
- Write short notes on harbour, tunnel, airport and railway engineering.
- Sources of water; quality parameters & drinking-water specifications.
- Hydrologic cycle; rainwater harvesting methods; dams and reservoirs.